

JEE ADVANCED - 1 | PAPER - 2

JEE 2022

Date	Maximum Marks	Duration
10th January, 2021	198	3 Hours

PAPER SCHEME

- The question paper consists of 3 Subject (Subject I: **Physics**, Subject II: **Chemistry**, Subject III: **Mathematics**). Each Part has **three** sections (Section 1, Section 2 & Section 3).
- Section 1** contains **SIX (06)** questions. The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 TO 9. **BOTH INCLUSIVE**.

Section 2 contains **6 Multiple Correct Answers Type Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

Section 3 contains **6 Numerical Value Type Questions**. The answer to each of the question is a numerical value. For each question, enter the correct numerical value of the answer. **If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled. (Example: 6, 81, 1.50, 3.25)**
- For answering a question, an ANSWER SHEET (OMR SHEET) is provided separately. Please fill your **Test Code**, **Roll No.** and **Group** properly in the space given in the ANSWER SHEET.

Name of the Candidate (In CAPITALS) :

Roll Number :

OMR Bar Code Number :

Candidate's Signature : Invigilator's Signature

MARKING SCHEME

SECTION – 1 (Maximum Marks: 18)	
➤	This section contains SIX (06) questions. The answer to each question is a SINGLE DIGIT INTEGER ranging from 0 TO 9. BOTH INCLUSIVE . For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
➤	Answer to each question will be evaluated according to the following marking scheme:
Full Marks :	+3 If ONLY the correct integer is entered;
Zero Marks :	0 If the question is unanswered:
Negative Marks:	–1 In all other cases.

SECTION - 2 (Maximum Marks: 24)	
➤	This section consists of SIX (06) Questions. Each question has FOUR options. ONE OR MORE THAN ONE of these four option(s) is(are) correct answer(s).
➤	Answer to each question will be evaluated according to the following marking scheme:
Full Marks:	+4 If only (all) the correct option(s) is(are) chosen
Partial Marks:	+3 If all the four options are correct but ONLY three options are chosen
Partial Marks:	+2 If three or more options are correct but ONLY two options are chosen and both of which are correct
Partial Marks:	+1 If two or more options are correct but ONLY one option is chosen, and it is a correct option
Zero Mark:	0 if none of the options is chosen (i.e. the question is unanswered)
Negative Marks:	–2 In all other cases.

SECTION – 3 (Maximum Marks: 24)	
➤	This section has SIX (06) Questions. The answer to each question is a NUMERICAL VALUE .
➤	For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled. (Example: 6, 81, 1.50, 3.25)
➤	Answer to each question will be evaluated according to the following marking scheme:
Full Marks:	+4 If ONLY the correct numerical value is entered;
Zero Marks:	0 In all other cases.

SUBJECT I : PHYSICS

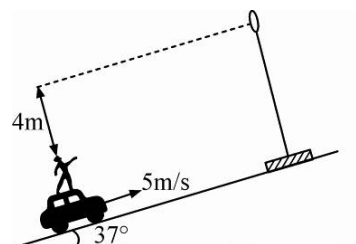
66 MARKS

SECTION 1

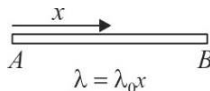
SINGLE DIGIT INTEGER

This section consists of SIX (06) questions. The answer to each question is a SINGLE DIGIT INTEGER ranging from 0 TO 9. BOTH INCLUSIVE.

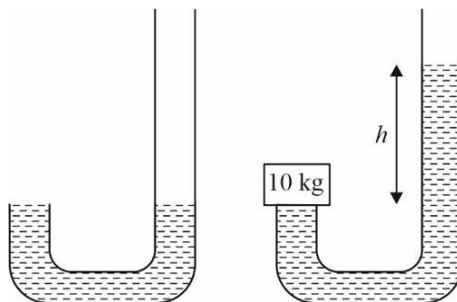
1. A man is travelling on a cart which is moving up an inclined plane at a speed of 5 m/s as shown. He throws a ball towards a stationary hoop located perpendicular to the incline in such a way that the ball moves parallel to the slope of incline while going through the centre of hoop which is 4 m high from the man's hand. What is the time (in seconds) taken by the ball to reach the centre of hoop? ($g = 10\text{ m/s}^2$)



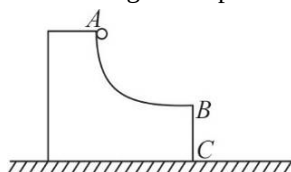
2. A thin rod AB of length l has non uniform mass distribution such that its mass per unit length varies with distance x from end A as $\lambda = \lambda_0 x$, where λ_0 is a positive constant of appropriate dimensions. If it is hinged at end A and a force F is applied normal to the rod at end B, the angular acceleration of the rod is α_A . However if it is hinged at end B and the force F is applied normal to the rod at end A, the angular acceleration of the rod is α_B . Find $\frac{\alpha_B}{\alpha_A}$. (Assume gravity free space)



3. A geostationary satellite is orbiting the earth at a height $6R$ above the earth's surface, where R is radius of earth. The time period of another satellite at a height $0.75 R$ from earth's surface would be _____ (in hours).
4. A J-shaped tube open at both ends has a cross sectional area of 100 cm^2 . It contains water such that its left limb is completely filled. Now a block of mass 10 Kg is kept on top of the left limb and additional water is poured into the right limb. Find the minimum height (in m) of the additional water column poured so that the block gets lifted up. Assume that the right limb is of sufficient length. (Density of water = 1000 Kg/m^3)



5. A particle is released from the top of a fixed track AB which is a quarter circle of radius 4m . It slides down its smooth surface, leaves the track at B and then lands on the ground. If the height BC is 4m , find the distance (in m) of the point of landing of the particle from C .



6. A particle of mass 1 kg is moving in one dimension under the action of a force that delivers a constant power 4W to the particle. If initial speed of the particle is zero, find its speed (in m/s) after 8 seconds .

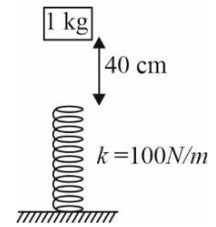
SECTION 2

MULTIPLE CORRECT ANSWERS TYPE

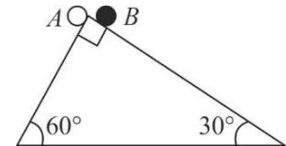
This section consists of SIX (06) Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

7. Two friends A and B are standing at a distance x apart in an open field and wind is blowing at a constant velocity u from A to B . A beats a drum and B hears its sound time t_1 after he sees the event. A and B interchange their position and A beats the drum again. This time B hears the drum time t_2 after he sees the event. Neglect the time light takes in travelling between the friends. Wind is continuously blowing in the same direction. Choose the correct options.
- (A) Velocity of sound in still air, $v = x \left(\frac{1}{t_1} + \frac{1}{t_2} \right)$
- (B) Velocity of wind, $u = \frac{x}{2} \left(\frac{1}{t_1} - \frac{1}{t_2} \right)$
- (C) Velocity of sound in still air, $v = \frac{x}{2} \left(\frac{1}{t_1} + \frac{1}{t_2} \right)$
- (D) Velocity of wind, $u = x \left(\frac{1}{t_1} - \frac{1}{t_2} \right)$
8. A syringe is filled with water. Its volume is $V_0 = 40\text{ cm}^3$ and cross section of its interior is $A = 8\text{ cm}^2$. The syringe is held vertically such that its nozzle is at the top, and its piston is pushed up with constant speed. Mass of the piston is $M = 50\text{ g}$ and the water is ejected at a speed of $u_0 = 2\text{ m/s}$. Cross section of the stream of water at the nozzle is $a = 2\text{ mm}^2$. Neglect friction, take the density of water to be $\rho = 10^3\text{ kg/m}^3$, $g = 10\text{ m/s}^2$. Choose the correct options.
- (A) The speed of the piston is 0.5 cm/s
- (B) The force required to push the piston decreases as the piston moves up
- (C) Total work done in emptying the syringe is nearly 0.115 J
- (D) Total work done in emptying the syringe is nearly 0.125 J

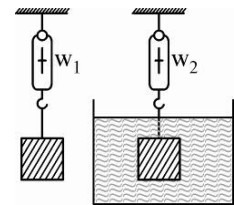
9. A long spring is kept vertical with its lower end attached to the ground and the upper end free. A block is released above the spring so that it strikes the top of spring after falling through a distance of 40 cm as shown. As the block compresses the spring, assume that the spring remains always vertical. Mass of the block is 1 kg and spring constant is 100 N/m. Choose the correct options. (Take $g = 10 \text{ m/s}^2$).



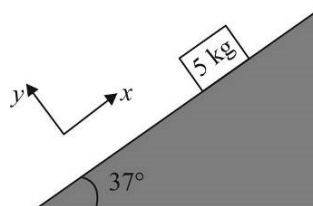
- (A) Velocity of the block is maximum just before striking the spring
 (B) Velocity of the block is maximum when the compression in spring is 10 cm
 (C) Maximum compression of the spring is 10 cm
 (D) Maximum compression of the spring is 40 cm
10. A hollow sphere A and a solid sphere B, both having same mass and radius are released simultaneously from the top of a right triangular fixed wedge as shown. They roll down the wedge without slipping. Choose the correct statements.



- (A) They reach the bottom of the wedge simultaneously
 (B) A reaches the bottom of wedge before B
 (C) They reach the bottom of wedge with the same kinetic energy
 (D) A reaches the bottom of wedge with speed more than that of B
11. A spring balance reads w_1 with a block suspended from it. If the block is submerged in a liquid whose density is less than density of block, the spring balance reads w_2 . The experiment is repeated twice, once on surface of earth and another time on surface of moon. If acceleration due to gravity on surface of moon is one-sixth of that on surface of earth, then choose the correct options.



- (A) w_1 is same on earth and moon
 (B) w_2 is same on earth and moon
 (C) $(w_1 - w_2)$ is same on earth and moon
 (D) $\frac{w_1}{w_2}$ is same on earth and moon
12. A block of mass 5 kg is kept on a fixed incline. The coefficient of friction between the block and incline is 0.5. x and y axis as shown are parallel to the incline and perpendicular to the incline respectively. Choose the correct options. (Take $g = 10 \text{ m/s}^2$, $\tan 37^\circ = 0.75$)



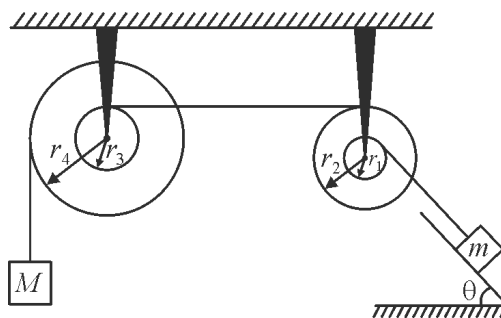
- (A) If the block is released, it slides down the incline with acceleration 6 m/s^2
 (B) If the block is released, it slides down the incline with acceleration 2 m/s^2
 (C) The minimum magnitude of force that needs to be applied on the block along $+x$ direction to keep the block at rest is 10N
 (D) The minimum magnitude of force that needs to be applied on the block along $+y$ direction to keep the block at rest is 20N

SECTION 3

NUMERICAL VALUE TYPE QUESTIONS

This section has SIX (06) Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled. (Example: 6, 81, 1.50, 3.25)

13. Starting from rest, a particle moves a distance x in a straight line with constant acceleration a . Then, the particle moves a further distance x along the same line at constant velocity. Finally, the particle, moving along the same line, decelerates uniformly at rate $\frac{a}{2}$ until it comes to rest. Let the maximum velocity of the particle during its motion be v_{MAX} and let its average velocity over the entire journey be v_{AVG} . The ratio $\frac{v_{MAX}}{v_{AVG}}$ is equal to _____.
14. A ball initially at rest is given a sharp horizontal impulse at a height h above the centre of the ball. As a result, the ball immediately begins to roll without slipping. If the ball is a solid sphere, the height $h = h_1$ while if the ball is a hollow sphere, the height $h = h_2$. Find $\frac{h_1}{h_2}$.
15. A thin hollow spherical shell of radius R is cut symmetrically and its lower half is removed to get a hemispherical shell. Now the open base is covered with a circular plate of radius R made of same material and having the same thickness as the shell. The centre of mass of this covered hemisphere lies at a distance of nR from the centre of base, where n is _____.
16. In the arrangement shown, the strings are massless, and the radii of the pulleys are: $r_1 = 3$ cm, $r_2 = 6$ cm, $r_3 = 4$ cm and $r_4 = 10$ cm. Friction is absent everywhere and the angle $\theta = \sin^{-1}\left(\frac{4}{5}\right)$. The value of $\frac{M}{m}$ such that the system remains in equilibrium is _____.



17. From a point on the ground at distance of $3m$ from the foot of a vertical wall, a ball is thrown at an angle 60° . The ball just clears the top of the wall and afterwards strikes the ground at a distance of 6 m from the foot of the wall on the other side. The height of the wall is ____ (in m). (Take $\sqrt{3} = 1.73$)
18. A block of mass 1 kg is kept at rest on a rough horizontal surface. The coefficient of friction between the block and surface is 0.5 . A horizontal force of 10 N is applied on the block for 6 seconds after which the direction of force is reversed keeping the magnitude same. Find the speed (in m/s) with which the block returns to its starting point. (Take $g = 10$ m/s², $\sqrt{3} = 1.732$)

SPACE FOR ROUGH WORK

SUBJECT II : CHEMISTRY

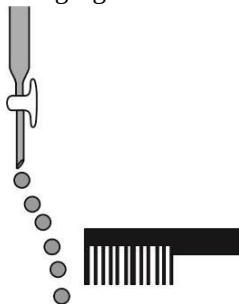
66 MARKS

SECTION 1

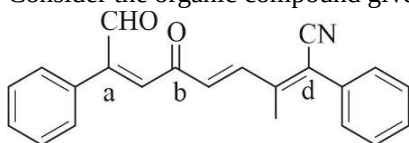
SINGLE DIGIT INTEGER

This section consists of SIX (06) questions. The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 TO 9. **BOTH INCLUSIVE.**

1. Consider the following compounds in the liquid form: Tetrachloromethane, Trichloromethane, Benzene, Chlorobenzene, p-Dihydroxy benzene, p-Xylene, Benzenamine and Ethanol. When a charged comb is brought near their flowing stream, how many of them show deflection as per the following figure?

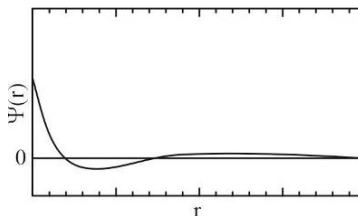


2. The valve on a cylinder containing initially 1 litre of an ideal gas at 7 atm and 25°C is opened to the atmosphere (temperature is 25°C). Assuming that the process is isothermal, how much magnitude of work (in atm-Litre) is done on the atmosphere by the action of expansion?
3. Consider the organic compound given below:



Where a, b and d are locants of carbon atoms in numbering of longest continuous principal carbon chain for systematic IUPAC naming of this compounds. Find the value of $(d + b - a)$.

4. How many electrons in a ground-state As atom in the gas phase have quantum numbers $n = 3$ and $l = 1$? [Given: Atomic Number of As = 33]
5. The formation of tetramminecopper(II) ion, $\text{Cu}(\text{NH}_3)_4^{2+}$, is a reversible reaction having formation constant $K_f = 1.1 \times 10^{13}$. The minimum concentration of free ammonia in solution required to ensure that at least 99% of the dissolved copper(II) ion is found in the form of its ammonia complex is 1.73×10^{-p} M. What is value of p? [Given: $\sqrt{3} = 1.732$]
6. An orbital has the radial wavefunction shown below. What is the value of the principal quantum number of the orbital?



SECTION 2

MULTIPLE CORRECT ANSWERS TYPE

This section consists of SIX (06) Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

7. The equation that is/are true for mixing of two ideal gases at constant temperature and pressure is:
(A) $\Delta U_{\text{mix}} = 0$ **(B)** $\Delta S_{\text{mix}} = 0$ **(C)** $\Delta H_{\text{mix}} = 0$ **(D)** $q_{\text{mix}} = 0$
8. Intensive variable(s) is/are:
(A) temperature **(B)** volume **(C)** pressure **(D)** density
9. Which of the following is/are INCORRECT for organic compound allene ($\text{CH}_2=\text{C}=\text{CH}_2$)?
(A) It is planar compound
(B) Nodal plane of both the pi bonds are lying in the same plane
(C) Maximum 5 atoms are lying along the straight line
(D) Percent s character in hybrid orbital of all carbon atoms is same.
10. The molecule possessing zero dipole moment is/are:
(A) CCl_4 **(B)** XeF_4 **(C)** SF_4 **(D)** PCl_5
11. Among the following pair of diatomic molecules, choose the one where the bonding $\sigma(2p)$ orbitals is higher in energy than the $\pi(2p)$ orbitals.
(A) O_2 and F_2 **(B)** N_2 and B_2 **(C)** C_2 and HF **(D)** C_2 and B_2
12. For which of the following reactions would the yield of products at equilibrium NOT increase on increasing pressure?
(A) $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g})$
(B) $\text{Ti}(\text{s}) + 2\text{Cl}_2(\text{g}) \rightleftharpoons \text{TiCl}_4(\text{g})$
(C) $2\text{C}_2\text{H}_4(\text{g}) + 2\text{H}_2\text{O}(\text{g}) \rightleftharpoons 2\text{C}_2\text{H}_6(\text{g}) + \text{O}_2(\text{g})$
(D) $4\text{HCl}(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{H}_2\text{O}(\text{l}) + 2\text{Cl}_2(\text{g})$

SPACE FOR ROUGH WORK

SECTION 3

NUMERICAL VALUE TYPE QUESTIONS

This section has SIX (06) Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25)

13. If **Z1** is the number of electrons in g sub shell, **Z2** is the azimuthal quantum number of an atomic orbital of g sub shell, **Z3** is the minimum principal quantum number of an atomic orbital of g sub shell, **Z4** is the number of angular nodes in an atomic orbital of the g sub shell and **Z5** is the number of elements expected in the g block of the periodic table then find the value of **Z1 + Z2 + Z3 + Z4 + Z5**.
14. A mixture of 1.20 mol of X, 2.10 mol of Y, and 0.950 mol of Z is found at equilibrium in a 1 litre vessel as shown below.
$$X + Y \rightleftharpoons Z$$

What is the value of the equilibrium constant, Kc?
15. A solution of a mixture of CaCl_2 and NaCl weighing 4.44 g was treated with sodium carbonate solution to precipitate all the calcium ions as calcium carbonate. The calcium carbonate so obtained was heated strongly and 0.56 g of CaO was obtained. The percentage of NaCl in the mixture is:
[Given: Atomic Mass of Na = 23, Cl = 35.5, Ca = 40 g/ Mol]
16. The mass (in kg) of photon with wavelength $1\text{\AA}$ is $p \times 10^{-32}$.
What is numerical value of p? [Given: $h = 6.626 \times 10^{-34} \text{ J s}$, $c = 3 \times 10^8 \text{ m/s}$]
17. The dipole moment of BrF is 1.35 D, and its bond length is 180 pm. What is the percent ionic character of the Br – F bond?
[Given: $1\text{D} = 3.34 \times 10^{-30} \text{ C m}$, Charge of electron = $1.6 \times 10^{-19} \text{ C}$]
18. A certain mass of gas occupies 5.50 liters at 300 K and 650 Torr. What will be its volume (liters) if it is cooled at 283 K and its pressure is increased to 980 Torr?

SPACE FOR ROUGH WORK

SUBJECT III : MATHEMATICS**66 MARKS****SECTION 1****SINGLE DIGIT INTEGER**

This section consists of **SIX (06)** questions. The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 TO 9. **BOTH INCLUSIVE.**

1. Let $A(0, 1)$, B, C be three points on $\frac{x^2}{4} + \frac{y^2}{1} = 1$, be vertices of an equilateral triangle, such that altitude from A on BC is $x = 0$. If length of side of this equilateral triangle be k , then $\frac{13k}{\sqrt{12}}$ is _____.
2. Two complex numbers z_1, z_2 are such that $z_1 \cdot z_2$ is purely imaginary & $\frac{z_1}{z_2}$ is purely real. Number of pairs (z_1, z_2) such that $|z_1| = |z_2| = 1$ is _____.
3. Let $\alpha = \cos \frac{2\pi}{20} + i \sin \frac{2\pi}{20}$, then value of $\sum_{r=0}^{19} \alpha^{10r}$ is _____.
4. Complete values of ' k ' for which the point $(k, 1)$ lies inside the major segment formed by circle $x^2 + y^2 - 3x + 1 = 0$ & the line $2x - y = 2$, is (a, b) , then value of ab is _____.
5. If one of the three normals drawn from the point $P(8, -2)$ to parabola $y^2 = 4ax$ ($0 < a < 3$) passes through $(3a, 0)$, then length of the latus rectum of parabola is _____.
6. Consider the parabola $y^2 = x - 4y + 3$ its focus is at (x_1, y_1) & vertex is at (x_2, y_2) , then $4(|x_2 - y_1| - |x_1 - y_2|)$ is _____.

SPACE FOR ROUGH WORK

SECTION 2

MULTIPLE CORRECT ANSWERS TYPE

This section consists of SIX (06) Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

7. Let C_1 & C_2 be centres of 2 circles whose radii are 2 & 4 respectively. Also $C_1C_2 = 10$ & direct common tangents of these circles touch them at P, Q, R, S . Another circle of radius ' r ' is drawn passing through P, Q, R, S , then:
 - (A) mid-point of C_1C_2 is centre of circle passing through P, Q, R, S
 - (B) centre of circle passing through P, Q, R, S divides C_1C_2 in 1 : 2
 - (C) $r = \sqrt{33}$
 - (D) $r = \sqrt{35}$
8. If three distinct normals can be drawn to the curve $y^2 = x$ from the point $(k, 0)$, then k can be:
 - (A) $\frac{1}{4}$
 - (B) $\frac{1}{2}$
 - (C) 1
 - (D) 2
9. If the line $4x + 2y + 1 = 0$ intersects the parabola $y^2 = -x$ at A & B and O is origin, then:
 - (A) $\tan(\angle AOB) = \left(\frac{2\sqrt{5}}{3}\right)$
 - (B) $\tan(\angle AOB) = -\frac{2\sqrt{5}}{3}$
 - (C) Circle described on AB as diameter touches line $4x - 1 = 0$
 - (D) tangents drawn at A & B to the parabola are not perpendicular
10. Consider the ellipse $\frac{x^2}{9} + \frac{y^2}{5} = 1$, then which of the following is(are) TRUE?
 - (A) Area of parallelogram formed by tangents at end points of latus rectum is 27 sq.units.
 - (B) Area of the parallelogram formed by tangents at end points of latus rectum is $\frac{27}{2}$ sq.units
 - (C) Area of rectangle formed by vertices of end points of latus rectum is $\frac{20}{3}$ sq.units.
 - (D) Area of parallelogram formed by end points of major & minor axis is $6\sqrt{5}$ sq. units.
11. Let $w \neq 1$ be a cube root of unity and a, b, c are integers then minimum value of $|a + bw + cw^2|^2$ is:
 - (A) 2 if a, b, c are three distinct integers
 - (B) 3 if a, b, c are three distinct integers
 - (C) 1 if a, b, c are not all equal
 - (D) 12 if minimum absolute difference between any two of a, b or c is 2

12. If $y = mx + c$ is common tangent to $\frac{x^2}{25} + \frac{y^2}{4} = 1$, $x^2 + y^2 = 16$ and $y^2 = 4ax$ ($a > 0$), then which of the following can be TRUE?
- (A) focus of the parabola is at $\left(\frac{8\sqrt{7}}{3}, 0\right)$ (B) $\sqrt{3}(m+c) = 2 + 4\sqrt{7}$
- (C) $\sqrt{3}(m+c) = -2 - 4\sqrt{7}$ (D) Latus rectum of the parabola is $\frac{2\sqrt{7}}{3}$

SPACE FOR ROUGH WORK

SECTION 3

NUMERICAL VALUE TYPE QUESTIONS

This section has SIX (06) Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled. (Example: 6, 81, 1.50, 3.25)

13. The focus of the parabola is at $\left(-\frac{2}{5}, -2\right)$ & its directrix is the common tangent to the circles $x^2 + y^2 - 10x - 24y + 120 = 0$ & $x^2 + y^2 = 400$, then length of the latus rectum of parabola is _____.
14. Area of the circle orthogonal to each of the circles $x^2 + y^2 - 6|x| - 6|y| + 14 = 0$ is λ , then $[\lambda]$ is _____. ($[\cdot]$ denotes greatest integer function)
15. Let $z = \frac{\left(\cos\left(\frac{\pi}{12}\right) + i\sin\left(\frac{\pi}{12}\right)\right)\left(\sin\frac{\pi}{3} - i\cos\frac{\pi}{3}\right)}{\left(\sin\frac{\pi}{4} + i\cos\frac{\pi}{4}\right)\left(\cos\frac{\pi}{6} - i\sin\frac{\pi}{6}\right)}$, then $\operatorname{Re}(z) - \operatorname{Im}(z)$ is _____.
16. Distance between the points of contacts of tangents to the ellipse $\frac{x^2}{16} + \frac{y^2}{1} = 1$ which are inclined at 60° with positive direction of x-axis is $\frac{\sqrt{769}}{\lambda}$, then λ is _____.
17. Equation of circle touching the line $2x - y - 1 = 0$ at $(3, 5)$ & having centre lying on $x + y - 5 = 0$ is $x^2 + y^2 + 2gx + 2fy + c = 0$, then $\frac{c}{|g+f|}$ is _____.
18. If x_1, x_2, x_3, x_4 are 4 distinct possible values of $\alpha - \beta$ where $\alpha^2 = 5 + 12i$ & $\beta^2 = 5 - 12i$, then $\sum_{i=1}^4 x_i^2$ is _____.

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